

Project: On The Grid

Avangrid Hackathon 2024: By the Brownies
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The challenge

Avangrid's renewable assets are nearing the end of their PPAs. Once they expire, these projects face volatile market prices, shifting demand, and uncertain generation. Estimate future revenues and decide whether to stay merchant or secure a fixed price that balances risk and reward.



Our Goal

We built a simple, transparent framework that helps answer the question

“At what price should we re contract these assets so that we’re protected from downside risk but still capture upside potential?”

Our model combines real data, machine learning, and risk analysis to:

- Predict future energy prices and revenues for each asset.
- Quantify risk using P50 (expected) and P75 (conservative) scenarios.
- Recommend fixed prices that align with Avangrid’s 75% risk-confidence level.

Methodology Overview

Step	Method	Output
1. Data Preprocessing	78,912 hourly records (3 yrs) → monthly P/OP	Clean dataset
2. Model Training	Gradient Boosting Regressor (GBR)	$R^2 = 0.95$ / $MAE = \$0.84$
3. Forecasting	Apply model to forward prices (2026–2030)	5-yr expected prices
4. Risk Analysis	1,000 Monte Carlo simulations ($\sigma = 15\%$)	P50 / P75 metrics



Model Performance and Validation

We trained a Gradient Boosting Regressor on 2022–2023 data and tested it on 2024 data to predict realized market prices using generation, market, month, and price features.

Model Accuracy:

Metric	Value	Meaning
R^2	0.951	Explains 95% of price variation, excellent predictive fit.
MAE	\$0.84 / MWh	Average error < \$1/MWh, highly accurate forecasts.
RMSE	\$2.83 / MWh	Low volatility in errors, large misses are rare.



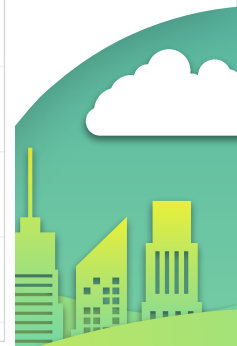
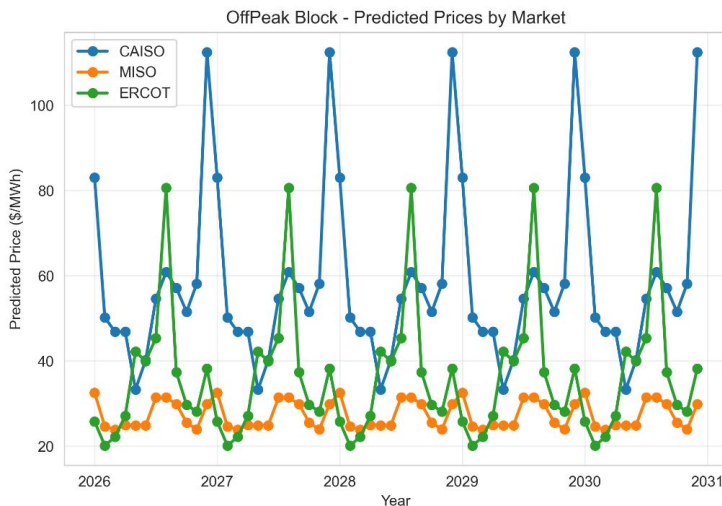
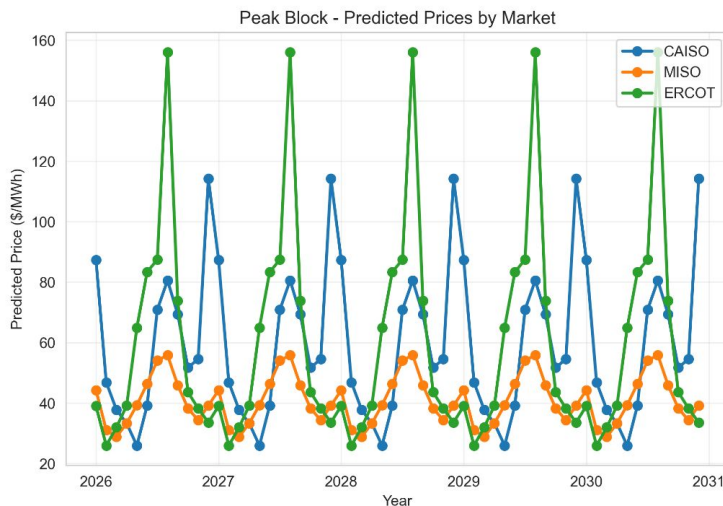
Market Price Forecasts

The model forecasts realistic price patterns across all markets, capturing seasonal highs in summer and lower winter prices.

ERCOT: Highest prices (~\$55/MWh Peak) strongest merchant upside.

CAISO: Moderate peaks, event-driven volatility.

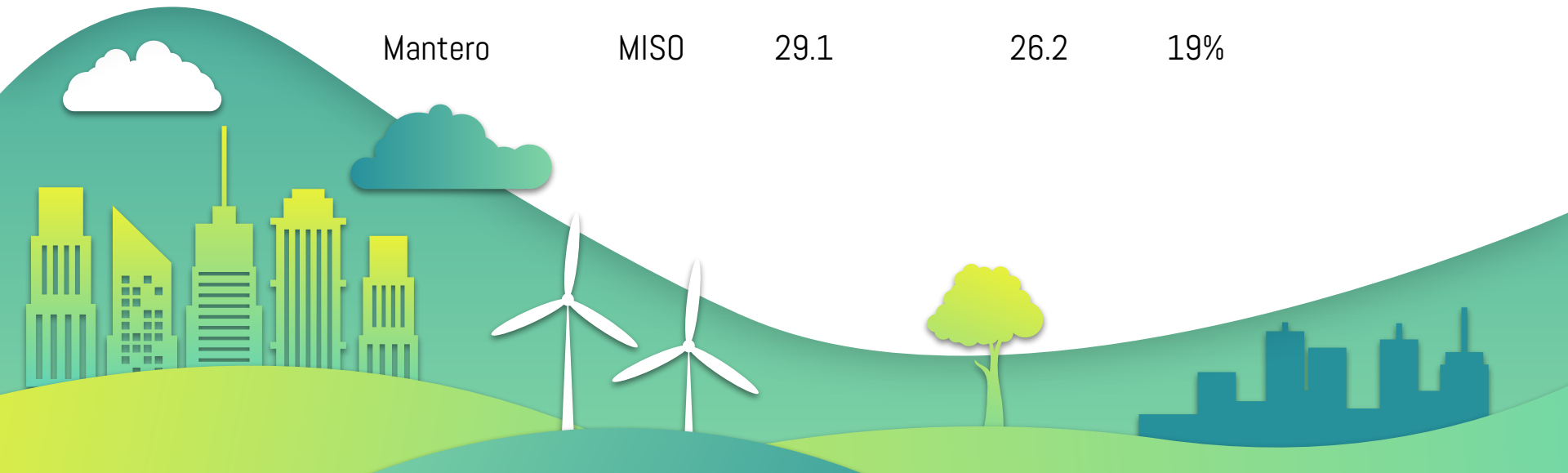
MISO: Most stable and lowest prices, best for staying merchant.



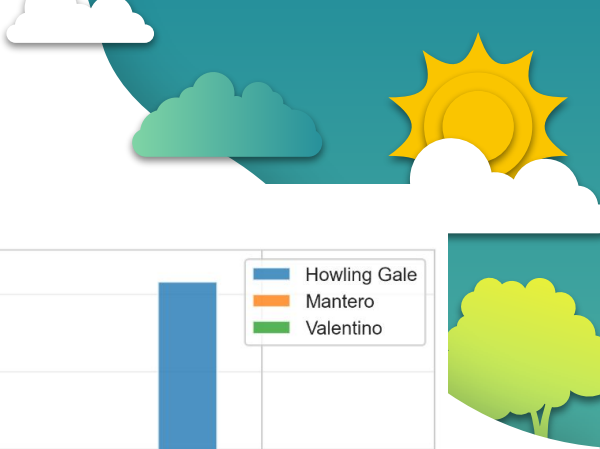
Revenue Forecast

Revenue is steady year-over-year, showing predictable generation and pricing patterns.

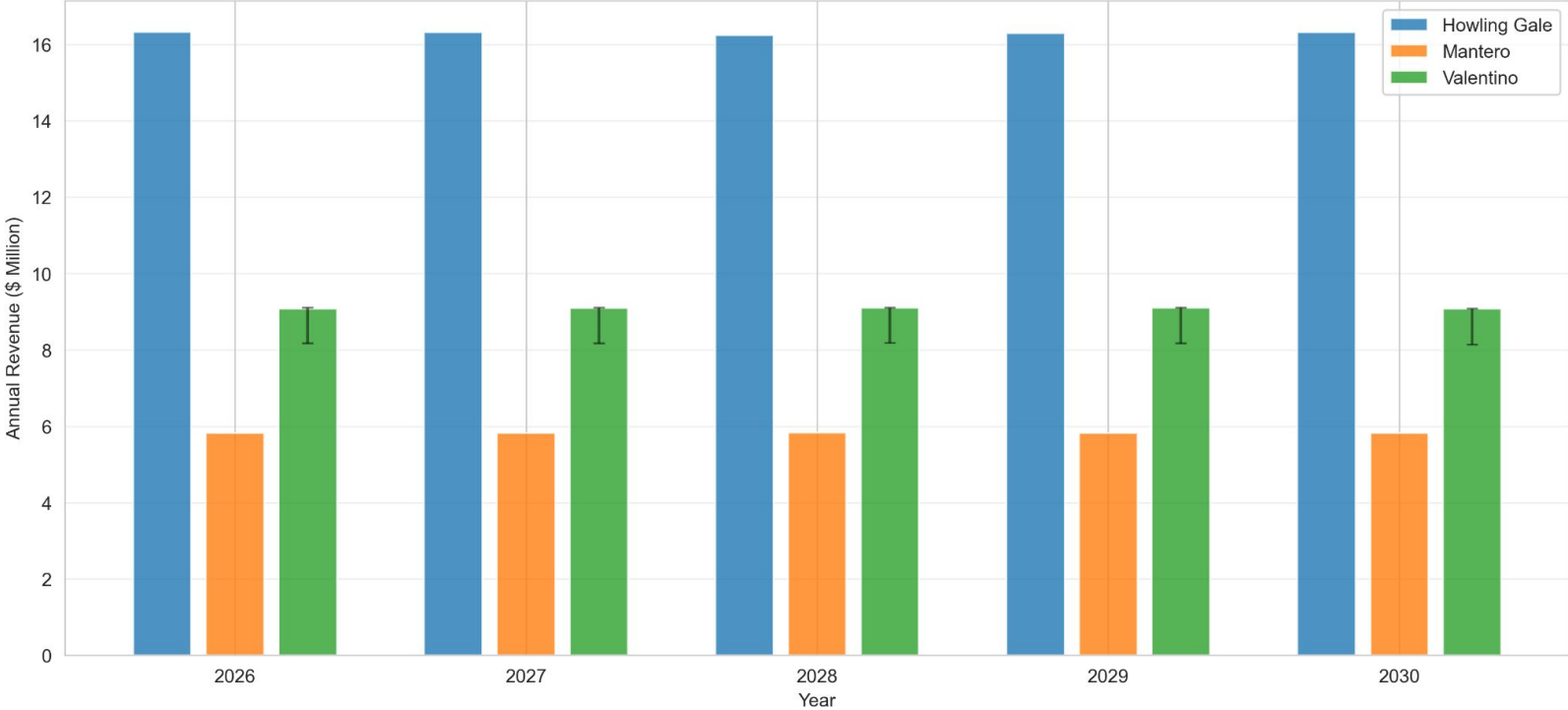
Asset	Market	Expected (\$M)	P75 (\$M)	Share
Howling Gale	CAISO	81.5	73.2	52%
Valentino	ERCOT	45.5	40.8	29%
Mantero	MISO	29.1	26.2	19%



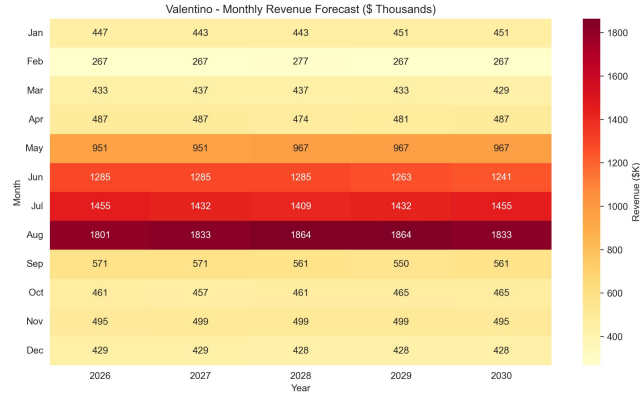
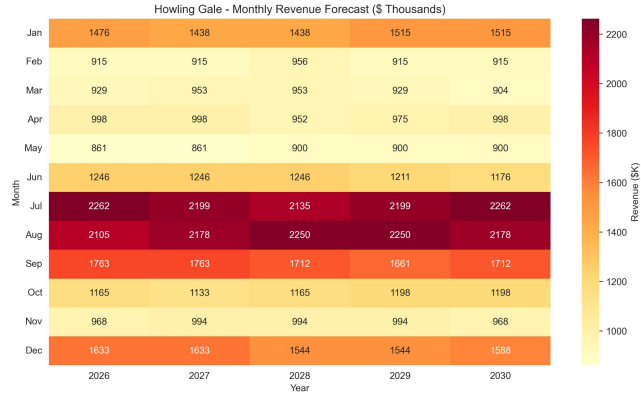
Revenue Forecast continued



Projected Annual Revenue by Asset (2026-2030)
with P50/P75 Risk Bands



Each heatmap shows monthly revenue



Risk and Sensitivity Analysis

Ran 1,000 Monte Carlo simulations to model market uncertainty.

Calculated P50 (median) and P75 (conservative) revenue outcomes.

Key Results

P50: \$156M → Expected median case.

P75: \$140M → 75% confidence floor for fixed-price deals.

Sensitivity: $\pm 10\%$ price change → $\pm \$15\text{M}$ revenue swing (linear).

Decision Example

A \$52/MWh fixed PPA $\approx$ \$145M (beats P75, but trades \$11M of upside).

Balanced strategy: counter at \$54–55/MWh to share risk and reward.



Recommended Fixed Prices

Asset	Market	P75 \$/MWh	Recommendation
Howling Gale	CAISO	52.3	Hybrid (50% fixed @ \$55)
Valentino	ERCOT	48.5	Lock 100% fixed @ \$50–52
Mantero	MISO	46.2	Stay merchant / >\$50 deal

Decision Rule:

Offer > P75 → Accept

P75 < Offer < P50 → Negotiate

Offer < P75 → Stay Merchant



DEMO

https://drive.google.com/drive/folders/1BKLpgRhL1GuUlygfCCVdvxnZ0-N_f9m-?usp=sharing

